

AI/ML intern DAY 6

TASK — BODY MEASUREMENT AND OUTFIT RECOMMENDATION SYSTEM

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1. Introduction

This project focuses on developing an AI-powered Smart Fashion Recommendation System using Artificial Intelligence, Computer Vision, MediaPipe, OpenCV, and Dataset Processing.

The system analyses uploaded human images and performs:

- body landmark detection
- body posture analysis
- shoulder and waist estimation
- body angle detection
- clothing size prediction
- intelligent outfit recommendation

The AI system compares estimated body measurements with clothing dataset measurements to recommend suitable outfit sizes.

2. Objective

The objective of this project was to:

- detect human body landmarks
 - estimate body measurements
 - analyse body posture and body tilt
 - process fashion datasets
 - recommend suitable clothing sizes
 - improve AI-based virtual fitting systems
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3. Technologies Used

Technology	Purpose
Python	Programming
OpenCV	Image Processing
MediaPipe	Human Pose Detection
Pandas	CSV Dataset Analysis
Artificial Intelligence	Recommendation Logic
Computer Vision	Body Tracking

4. Features of the System

The AI system successfully implemented:

- body landmark detection
 - body position detection
 - body angle estimation
 - skeletal visualization
 - body measurement analysis
 - fashion dataset comparison
 - intelligent outfit recommendation
-

5. Workflow of the System

Step 1 — Human Image Input

The AI system accepts uploaded human body images for analysis.

Step 2 — Body Landmark Detection

MediaPipe Pose Estimation detects:

- shoulders
- elbows
- wrists
- hips
- knees

- ankles

using AI body tracking.

Step 3 — Body Measurement Estimation

The AI system estimates:

- shoulder width
- chest width
- waist width
- torso length
- sleeve length

using body landmark coordinate distances.

The distance between landmarks is calculated using:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

-10-8-6-4-2246810-10-5510A(6.0, 6.0)B(-6.0, -6.0)d = 16.97Delta x = 12Delta y = 12

Step 4 — Pixel to Centimeter Conversion

The user provides body height in centimeters.

The AI calculates scale factor using:

$$\text{Scale Factor} = \frac{\text{Real Height (cm)}}{\text{Image Height in Pixels}}$$

This converts body measurements from pixels into centimeters.

Step 5 — Position and Angle Detection

The AI system analyses:

- body posture
- shoulder alignment
- body tilt
- body orientation angle

for posture understanding.

Step 6 — Dataset Comparison

The system reads clothing size data from CSV dataset containing:

- clothing brands
- garment types
- chest measurements
- waist measurements
- fit categories

The estimated body measurements are compared with dataset values.

Step 7 — Outfit Recommendation

The AI system recommends:

- suitable outfit size
- clothing category
- matching fit type

based on closest measurement matching.

6. Code Implementation:

```
1  import cv2
2  import mediapipe as mp
3  import pandas as pd
4  import math
5
6  height = float(input("Enter Height (cm): "))
7  fit = input("Preferred Fit (Slim/Regular/Loose): ")
8
9  data = pd.read_csv("fashion_brand_measurements_dataset")
10
11 image = cv2.imread("woman.jpg")
12
13 if image is None:
14     print("Image not found")
15     exit()
16
17 image = cv2.resize(image, (720, 720))
18
19 mp_pose = mp.solutions.pose
20 mp_draw = mp.solutions.drawing_utils
21
22 pose = mp_pose.Pose(
23     static_image_mode=True,
24     min_detection_confidence=0.7
25 )
```

```

rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

results = pose.process(rgb)

if results.pose_landmarks:

    lm = results.pose_landmarks.landmark

    h, w, _ = image.shape

    ls = lm[11]
    rs = lm[12]

    x1, y1 = int(ls.x*w), int(ls.y*h)
    x2, y2 = int(rs.x*w), int(rs.y*h)

    shoulder_pixels = math.sqrt(
        (x2-x1)**2 + (y2-y1)**2
    )

    scale = height / h

    shoulder = shoulder_pixels * scale
    chest = shoulder * 1.8
    waist = chest * 0.85

```

```

angle = int(
    math.degrees(
        math.atan2(y2-y1, x2-x1)
    )
)

if abs(y1-y2) < 20:
    position = "Straight"
elif y1 > y2:
    position = "Tilted Right"
else:
    position = "Tilted Left"

match = data[
    (data['chest_cm'] >= chest-5) &
    (data['waist_cm'] >= waist-5) &
    (data['fit_type'].str.lower() == fit.lower())
]

recommendation = "No Match"

if not match.empty:

    best = match.iloc[0]

```

```

best = match.iloc[0]

recommendation = (
    f"{best['brand']} | "
    f"{best['garment_type']} | "
    f"{best['size']}"
)

mp_draw.draw_landmarks(
    image,
    results.pose_landmarks,
    mp_pose.POSE_CONNECTIONS
)

cv2.putText(
    image,
    f"Chest: {chest:.1f} cm",
    (20,40),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (0,255,255),
    2
)

```

```

cv2.putText(
    image,
    f"Waist: {waist:.1f} cm",
    (20,80),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (0,255,255),
    2
)

cv2.putText(
    image,
    f"Angle: {angle}",
    (20,120),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255,255,0),
    2
)

cv2.putText(
    image,
    f"Position: {position}",
    (20,160),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255,255,0),
    2
)

```

```

        0.7,
        (255,255,0),
        2
    )

    cv2.putText(
        image,
        recommendation,
        (20,200),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.6,
        (0,255,0),
        2
    )

    print("Recommended:", recommendation)

else:
    print("No Human Detected")

cv2.imwrite("output.jpg", image)

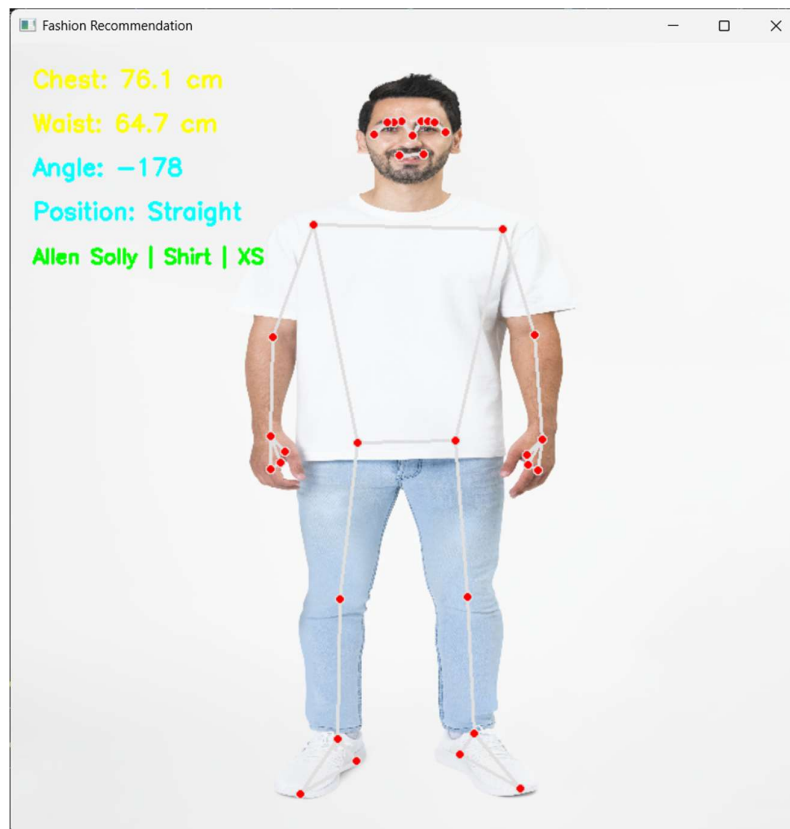
cv2.imshow("Fashion Recommendation", image)

cv2.waitKey(0)

cv2.destroyAllWindows()

```

Output:



7. Algorithm

Step 1

Load uploaded image using OpenCV.

Step 2

Detect human body landmarks using MediaPipe Pose.

Step 3

Calculate body landmark distances.

Step 4

Convert pixel distances into centimeters.

Step 5

Detect body position and body angle.

Step 6

Read fashion dataset from CSV file.

Step 7

Compare body measurements with clothing dataset.

Step 8

Recommend suitable outfit size and fit type.

8. Dataset Information

The fashion dataset contains:

- brand names
- garment categories
- chest measurements
- waist measurements
- clothing sizes
- fit categories

Example dataset fields:

Field	Description
brand	Clothing brand
garment_type	Type of clothing

size	Clothing size
chest_cm	Chest measurement
waist_cm	Waist measurement
fit_type	Clothing fitting type

9. Applications

The system can be used in:

- virtual fitting rooms
 - AI fashion assistants
 - online shopping systems
 - intelligent clothing recommendation
 - smart retail technology
 - digital fashion platforms
-

10. Advantages

The advantages include:

- automatic size prediction
 - intelligent outfit recommendation
 - improved shopping experience
 - reduced clothing return rates
 - AI-based body analysis
 - efficient virtual fitting
-

11. Challenges Faced

Challenge	Impact
Poor lighting	Reduces landmark accuracy
Side body posture	Affects measurement estimation
Loose clothing	Reduces body visibility
Complex background	Reduces detection quality

12. Expected Output

The AI system generates:

- body landmark visualization
 - skeletal tracking
 - body measurements
 - body angle detection
 - body posture analysis
 - intelligent outfit recommendation
 - clothing size prediction
-

13. Deliverables

The project deliverables include:

- AI body landmark detection system
 - body measurement estimation
 - body posture analysis
 - outfit recommendation system
 - CSV dataset processing
 - AI testing screenshots
 - output image generation
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14. Learning Outcomes

During this project, the following concepts were understood:

- AI body landmark tracking
 - body measurement estimation
 - skeletal visualization
 - body posture analysis
 - dataset-driven recommendation
 - computer vision workflows
 - AI fashion technology systems
-

15. Conclusion

The AI-Powered Body Measurement and Smart Outfit Recommendation System successfully demonstrated the integration of:

- Artificial Intelligence
- Computer Vision
- Human Pose Estimation
- Dataset Processing
- Intelligent Fashion Recommendation

The project improved understanding of:

- AI body tracking
- measurement estimation
- body posture analysis
- dataset comparison logic
- intelligent outfit recommendation systems

This project demonstrates how Artificial Intelligence can improve digital fashion systems and online shopping experiences using body analysis and intelligent recommendation workflows.